TT-F-340C
February 6, 1979
SUPERSEDING
Fed. Spec. TT-F-340B
December 7, 1973

FEDERAL SPECIFICATION

FILLER, WOOD, PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a plastic wood filler suitable for use with acetone solvent and for such repair work as building up or filling in wood patterns or joiner work.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gauge and Lighter.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge *from Business* Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(*Federal Government activities* may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

D 1729 - Visual Evaluation of Color Differences of Opaque Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Condition in container. When tested as specified in 4.4.1, there shall be no skinning, lumps, grit, hard or dry caking. There shall be no settling which cannot be readily re-mixed to a smooth, homogeneous state.

3.2 Hardenability. When tested as specified in 4.4.2, the filler shall not set-to-touch within 3 minutes, but shall dry hard within 4 hours.

3.3 Adhesion. When tested as specified in 4.4.3, not more than one-half of the material on a test panel shall be removed.

3.4 Cohesion. The filler shall have a tensile strength of not less than 300 lb/in^2 when tested as specified in 4.4.4.

3.5 Characteristics of the hardened material. When tested as specified in 4.4.5, the hardened filler shall be capable of being readily cut, sawed, bored, reamed, and filed, and shall not crack or split when nailed.

3.6 Stability of the hardened material. When tested as specified in 4.4.6, the hardened filler shall show no softening or swelling.

3.7 Color. Unless otherwise specified, the color of the filler shall be light, commercially designated as natural, and characteristic of the pigments used. If a definite color is required, such as walnut or mahogany, it shall be as specified (see 6.2).

3.8 Compatibility with acetone. When tested as specified in 4.4.8, the wood filler shall not settle out, but shall mix with the acetone.

3.9 Compatibility with lacquer topcoat. The filler shall not become gummy or lose adhesion from the substrate when tested as specified in 4.4.9.

3.10 Sag test. The filler, when tested as specified in 4.4.10, shall not sag more than 1/8 in (3.2 mm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

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4.2 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery and selected at random. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.3 Testing of the end item. For the purpose of sampling for tests, the lot shall be defined as the end product of all raw materials mixed, blended, or processed in a single operation. The lot shall be expressed in units of pounds. Samples from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an acceptable quality level (AQL) of 1.5.

4.4 Test procedures. The filler shall be tested according to the test methods indicated in table I. Unless otherwise specified, all tests shall be performed at standard testing conditions which are a temperature of 23 $\pm$ 1° C and a relative humidity of 50 $\pm$ 5 percent. All test reports shall contain the individual values utilized in expressing the final result. All tests shall be evaluated for conformance to the requirements specified in section 3. Failure to pass any test, or noncompliance with any requirement, shall be cause for rejection of the sample.

4.4.1 Condition in container. Before agitating the contents of the container, note whether skinning has occurred. Next, thoroughly stir the material and examine for compliance with the requirements in 3.1.

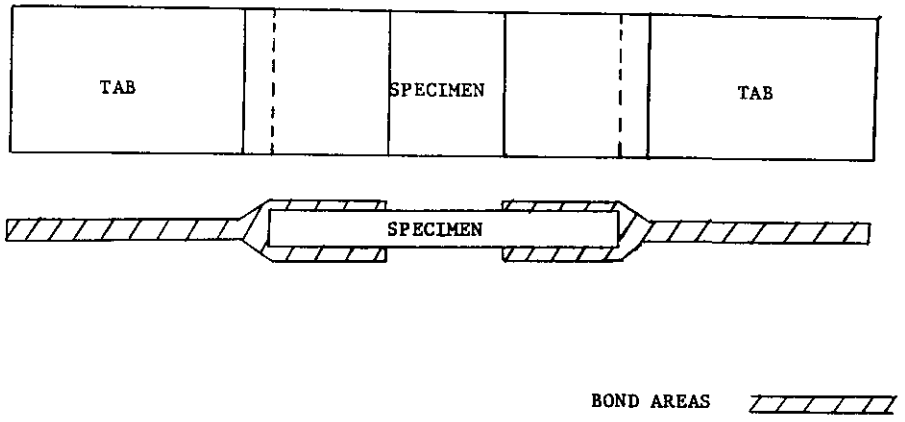
4.4.2 Hardenability. The filler shall be spread on a maple wood panel, previously cleaned with acetone, to a thickness of 1/16 in (2 mm) over an area of 6 in^2 (39 cm^2). The film shall be tested at points not less than 1/2 in (13 mm) from the edges. The set-to-touch time shall be determined by lightly touching the test film with the tip of the finger and immediately placing the finger against a piece of clean clear glass. For the purpose of this test, the pressure of the finger tip against the filler shall not be greater than that required to transfer a spot of the filler from 1/8 to 3/16 in in cross section. The film shall be considered set-to-touch when it shows a tacky condition but none of it adheres to the finger. Dry hard time shall be determined in the following manner. With the end of the thumb resting on the test film and the forefinger supporting the test panel, exert a maximum downward pressure (without twisting) of the thumb on the film. Lightly polish the contacted area with a soft cloth. The film shall be considered dry-hard when any mark left by the thumb is completely removed by the polishing operation. Examine the film for compliance with the requirements in 3.2.

4.4.3 Adhesion. The test panel shall be prepared as specified in 4.4.2. However, before applying the film of filler, weigh the test panel. Then, apply the filler and allow it to harden for 24 hours at standard conditions. After conditioning, weigh the panel with filler and determine the weight of the film. With the aid of a spatula or putty knife, pry the edge of the film from the test panel surface. Repeat the procedure at three points located 90 degrees from each other. Wipe away the loose filler and weight the panel with filler after the test. Determine the weight of filler lost during the test. Divide the weight of filler lost by the original weight of film of filler for compliance with 3.3.

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4.4.4 Cohesion. A briquet with a 0.25 in^2 (1.6 cm^2) cross section shall be prepared for the cohesion test, using the plastic wood filler as delivered. The briquet shall be prepared using 2 templates 1/8 in (3.2 mm) high, 3-3/4 in (95 mm) by 1 in (25 mm) inside, and approximately 4-3/4 in (121 mm) by 2 in (51 mm) outside. Form the briquet on a teflon plate or other release agent, in 2 successive layers, by placing one template on top of the other and using each template to form one layer. Allow 2 minutes between layers for solvent evaporation. After hardening for 1 hour, the briquet is released by carefully cutting around the inside edge of the mold with a razor blade. The mold is then removed. The briquet is allowed to harden at least 48 hours and pulled within 72 hours. The actual cross-sectional area of the briquet is determined in order that the lb/in^2 can later be calculated. Tabs shall be applied to both ends of the specimens covering 1 in^2 (6.5 cm^2) at each end. The tabs shall be made from cotton duck, having a weight of 20 - 30 oz/yd^2. The tabs shall extend 2 in (51 mm) beyond the specimen ends and be bonded together to provide a double thickness as shown in figure 1, to be held by the grips of the tensile testing machine. The cotton duck tabs shall be bonded to the specimen and to each other beyond the ends of the specimen with a polyvinyl acetate adhesive. A tensile load shall be applied to the test briquet at a rate of 2 in/min, until complete failure occurs. Evaluate for conformance with 3.4.

FIGURE 1
TABS FOR TENSILE TESTING



4.4.5 Characteristics of the hardened material. Broken halves of briquets from the testing in 4.4.4, shall be used to determine whether a knife which will readily cut white pine will cut the hardened plastic wood filler, and whether the plastic wood filler can be readily sawed, bored, reamed and filed. A 4-penny wire nail shall be driven through a part of the briquet to determine nailability. Observe for compliance with 3.5.

4.4.6 Stability of hardened material. Place a portion of a briquet, prepared as specified in 4.4.4, in a beaker containing sufficient tap water to completely submerge the sample. Place a weight on the sample to keep it submerged. Examine sample after soaking for 24 hours at standard conditions and evaluate for compliance with 3.6.

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4.4.7 Color. A panel of single thickness window glass, cleaned and coated as specified in 4.4.2, shall be illuminated and viewed as specified in ASTM D 1729 to determine compliance with 3.7.

4.4.8 Compatibility with acetone. When 4 parts by weight of filler are thinned with 1 part by weight of acetone, the resulting material shall comply with the requirement in 3.8.

4.4.9 Compatibility with lacquer topcoat. Prepare a test panel as specified in 4.4.2 and dry for 24 hours. Brush a coat of a lacquer-based paint over the dried wood filler. Allow 2 hours for the paint to dry and examine for compliance with 3.9.

4.4.10 Sag test. A cold-rolled steel test panel, measuring 6 x 10 in (152 x 254 mm), shall be cleaned to no water break. Prepare a 6 by 10 in (152 x 254 mm) template of 18-gauge steel with a 3 by 7 in (76 x 178 mm) rectangle cut from the center. The test panel shall have reference lines extending across the 6 in (152 mm) dimension 1.5 in (38 mm) from the top and bottom. These are for positioning the template. Another line shall be drawn similarly 1-3/8 in (35 mm) from the bottom. This is to measure sagging. The template is placed over the panel so the 3 by 7 in (76 x 178 mm) opening coincides with the reference lines. Apply a sample of well-mixed filler to the opening. Using another steel panel as a scraper, level-off the sample of filler to just fill the opening in the template. The scraper shall be held vertical to the template. Hold the template firmly to prevent slipping while scraping down the coating. Remove the template and place the panel at an angle of 30° to the horizontal. Examine for compliance with 3.10.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The filler shall be furnished in 4-ounce metal cans or one-pound metal cans, as specified (see 6.2). The cans shall conform to PPP-C-96, type V, class 2. Exterior Plan B coating shall be required.

5.1.2 Commercial. The filler shall be packaged in cans in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A. Ninety-six 4-ounce cans or forty-eight 1-pound cans of filler, packaged as specified in 5.1.1, shall be packed in a close-fitting box conforming to PPP-B-636, grades V3c, V3s or V2s. The boxes shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636. Alternatively, wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive.

5.2.2 Commercial. The filler, packaged as specified in 5.1, shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. This plastic wood filler can be used to fill holes and cracks in wood, and is suitable for such repair work as joiner work and building up wood patterns.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Color of wood filler required if other than natural (see 3.7).
(c) Whether 1/4- or 1-pound quantities are required (see 5.1.1 and 5.1.2).
(d) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 Shelf-life surveillance. For the purpose of shelf-life surveillance, the filler stored in the orginal, unopened container shall meet the requirements of paragraphs 3.1 and 3.4.

MILITARY INTEREST:

The DOD has waived coordination of this document until further notice

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITY:

NASA - JFK

Orders for this publication are to be placed with General Services Administration acting as an agent for Superintendent of Documents. See section 7 of this specification to obtain extra copies and other documents referenced herein.

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